

Software Development Technologies

BIS605

Software Development Technologies for Digital Business

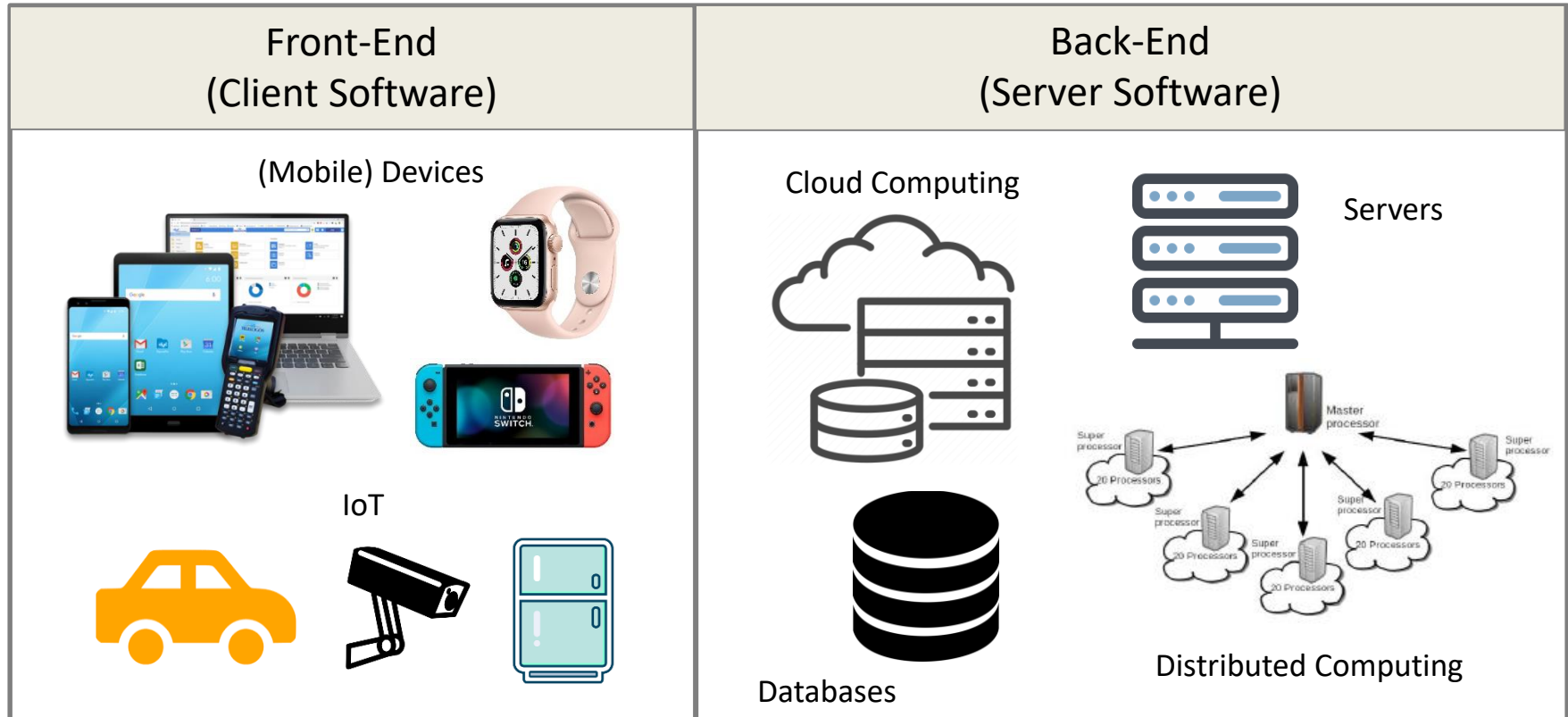
Asst. Prof. Dr. Worarat Krathu

References

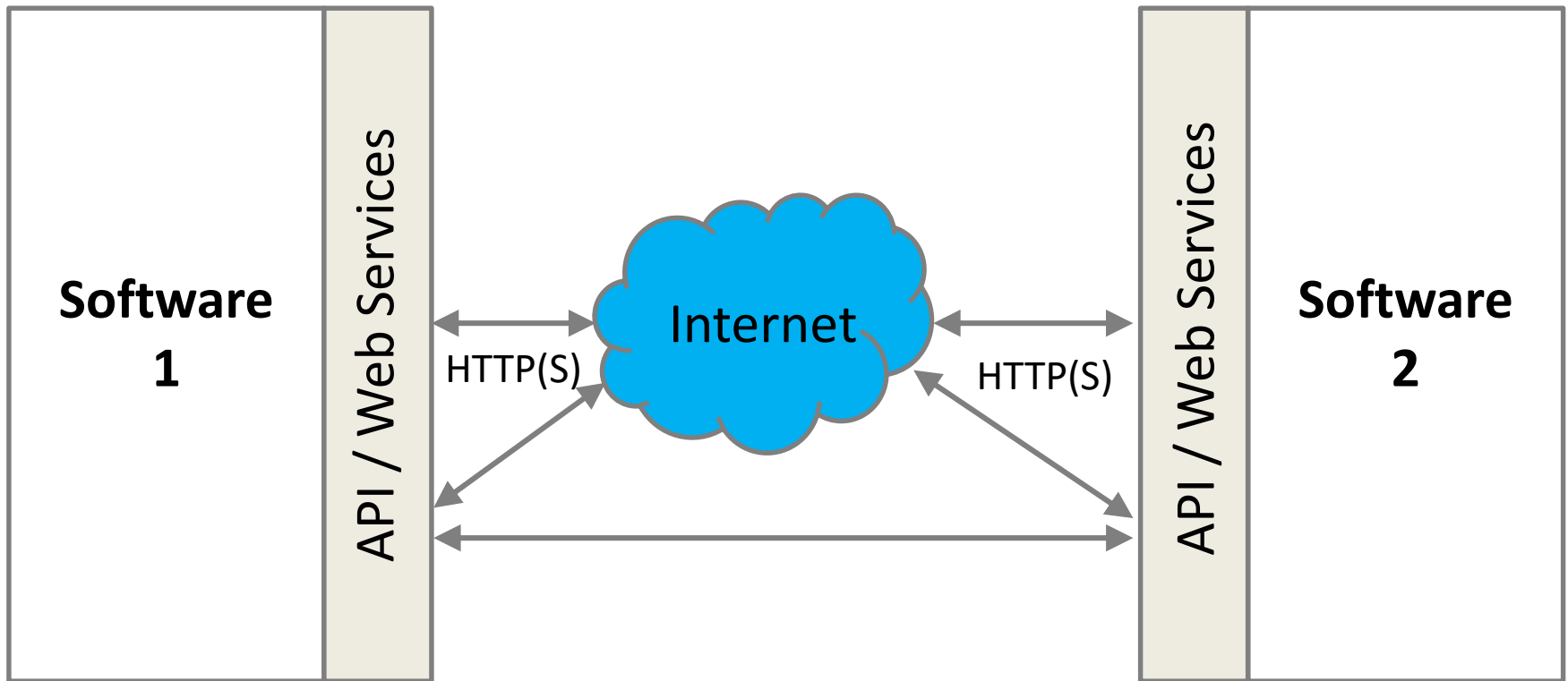
- https://www.w3schools.com/whatis/whatis_fullstack.asp
- <https://hackr.io/blog/web-development-ide>
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- <https://www.perforce.com/blog/vcs/what-source-control>
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- <https://blog.pjjop.org/docker-tutorial-for-beginners/>
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- <https://medium.com/dot-lab/testing-apis-using-postman-f879ffc4cc23>
- <https://skillcrush.com/blog/front-end-back-end-full-stack/>

Software and Related Components

Software



Software and Related Components



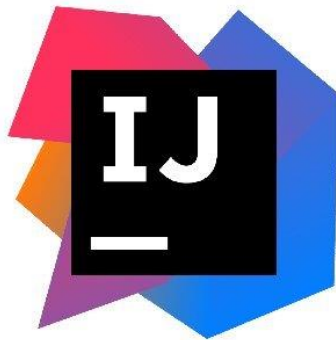
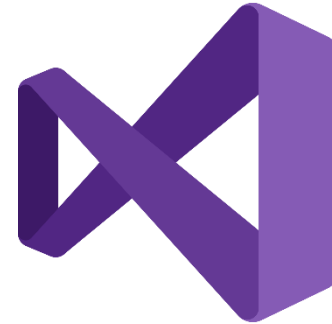
DEVELOPER TOOLS

IDE

- Integrated Development Environment (IDE) - software that consolidates the basic tools needed for software testing and writing
- Main features
 - Text editor
 - Compiler or Interpreter
 - Build or Make Integration
 - Debugger
 - Syntax Highlighter
 - Graphical User Interface (GUI)
 - Other possible features

Some Top IDE

- Visual Studio
- Netbeans
- IntelliJ IDEA
- PyCharm
- Eclipse



Source Control Tools

- A tool to track and manage changes to code to ensure that the developers are always working on the right version of source code
- Source Control vs. Version Control
 - Source control is specific to source code
 - Version control covers large binary files and digital assets
- Some source control tools
 - Git
 - Perforce Helix Core
 - Subversion
 - ClearCase

GIT / GitHub

- It is one popular source control tools
- Git is free and open source distributed version control system
- Some popular Git source control options include GitHub, GitLab, and Bitbucket

In Simple Terms

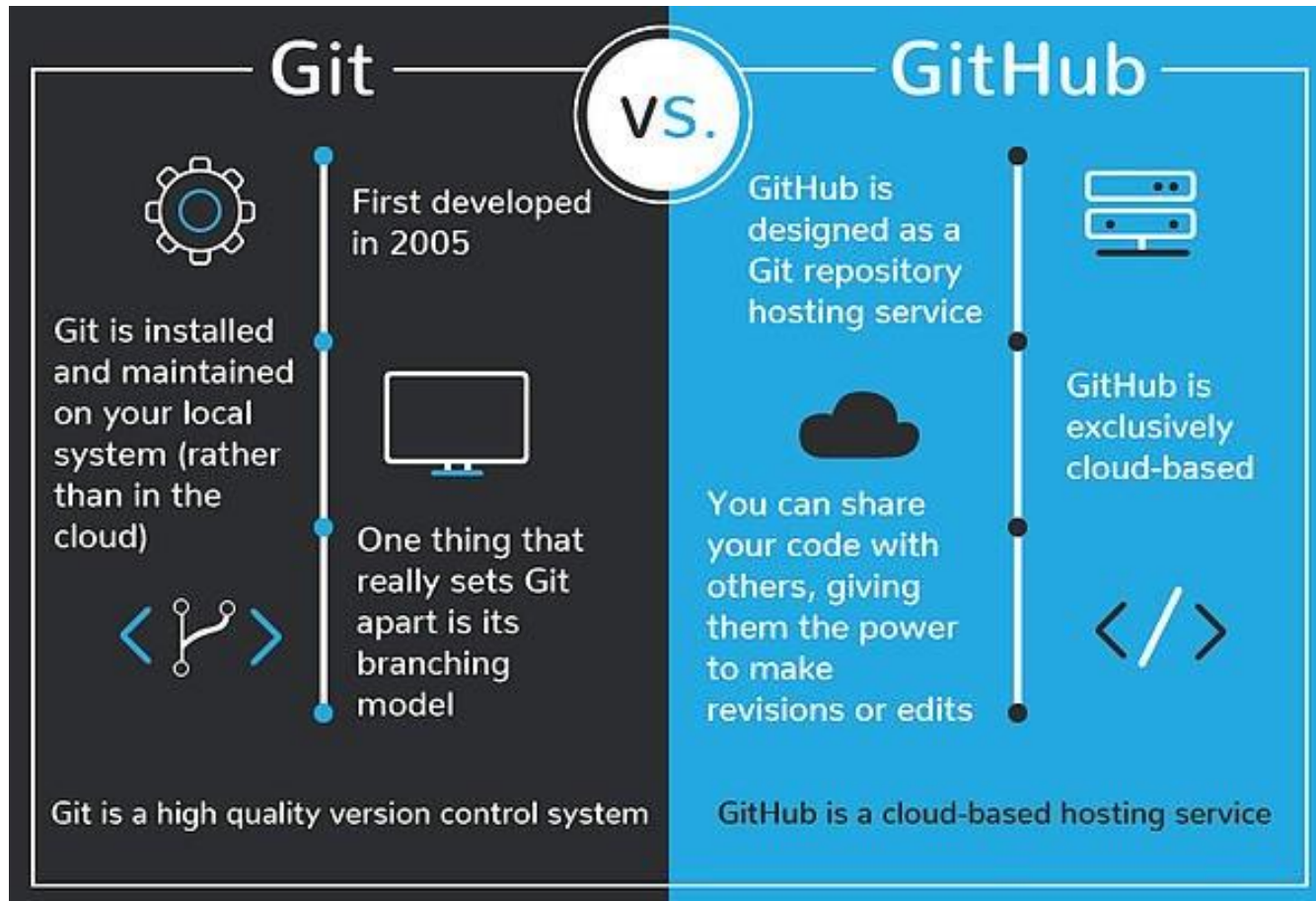
Git

is a version control system that lets you manage and keep track of your source code history

Git
Hub

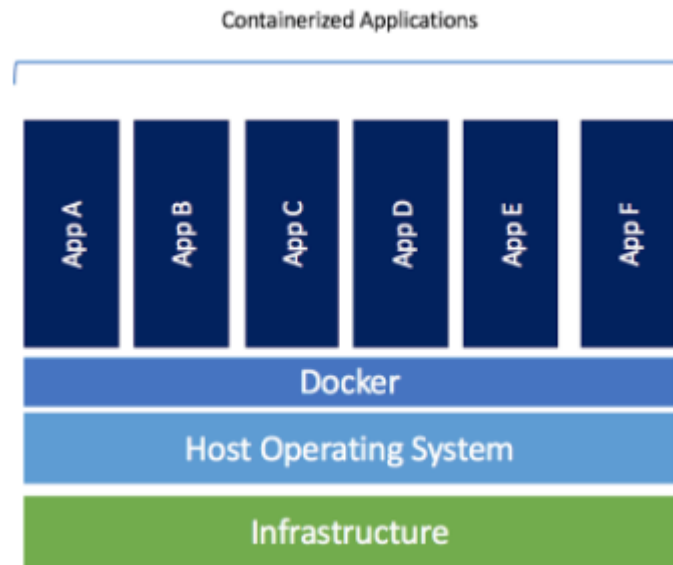
is a cloud-based hosting service that lets you manage Git repositories

GIT / GitHub

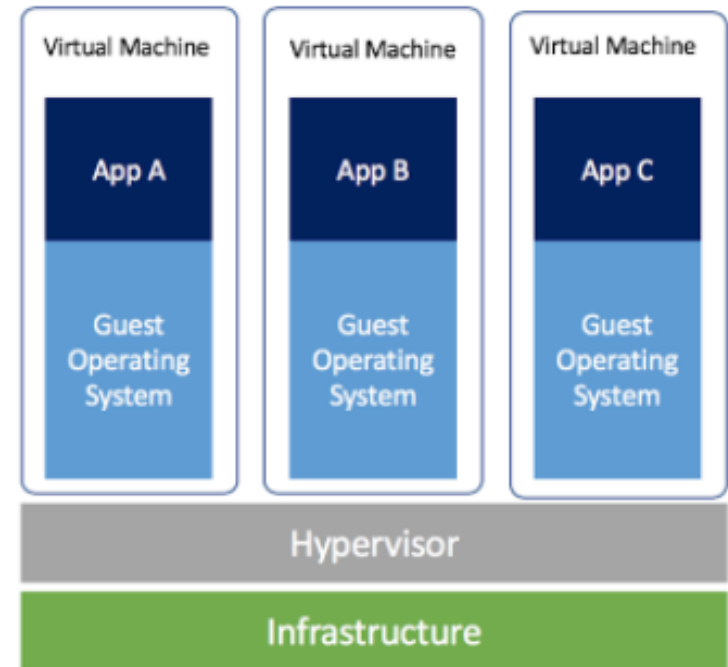
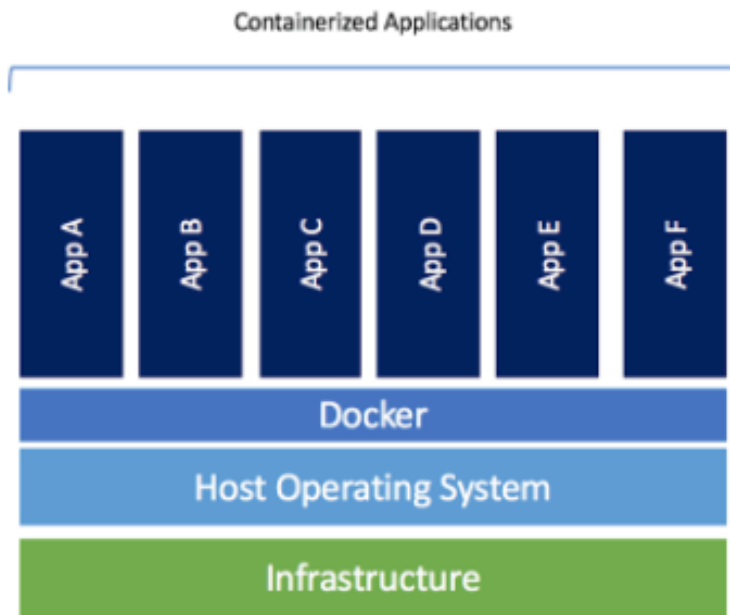


Docker

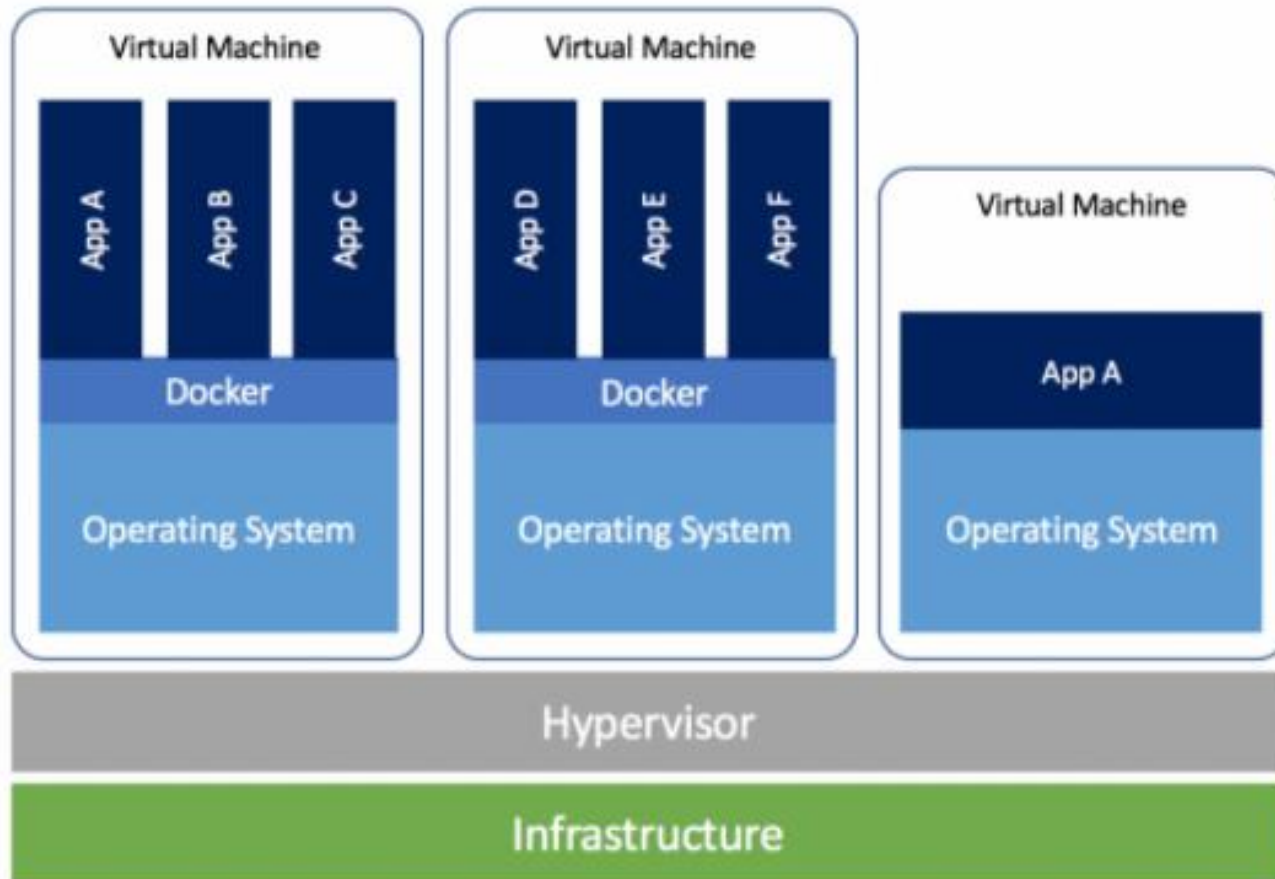
- A container platform to build, manage and secure applications
- Docker simplifies and accelerates workflow, while giving developers the freedom to innovate with their choice of tools, application stacks, and deployment environments for each project



Docker Vs. VM

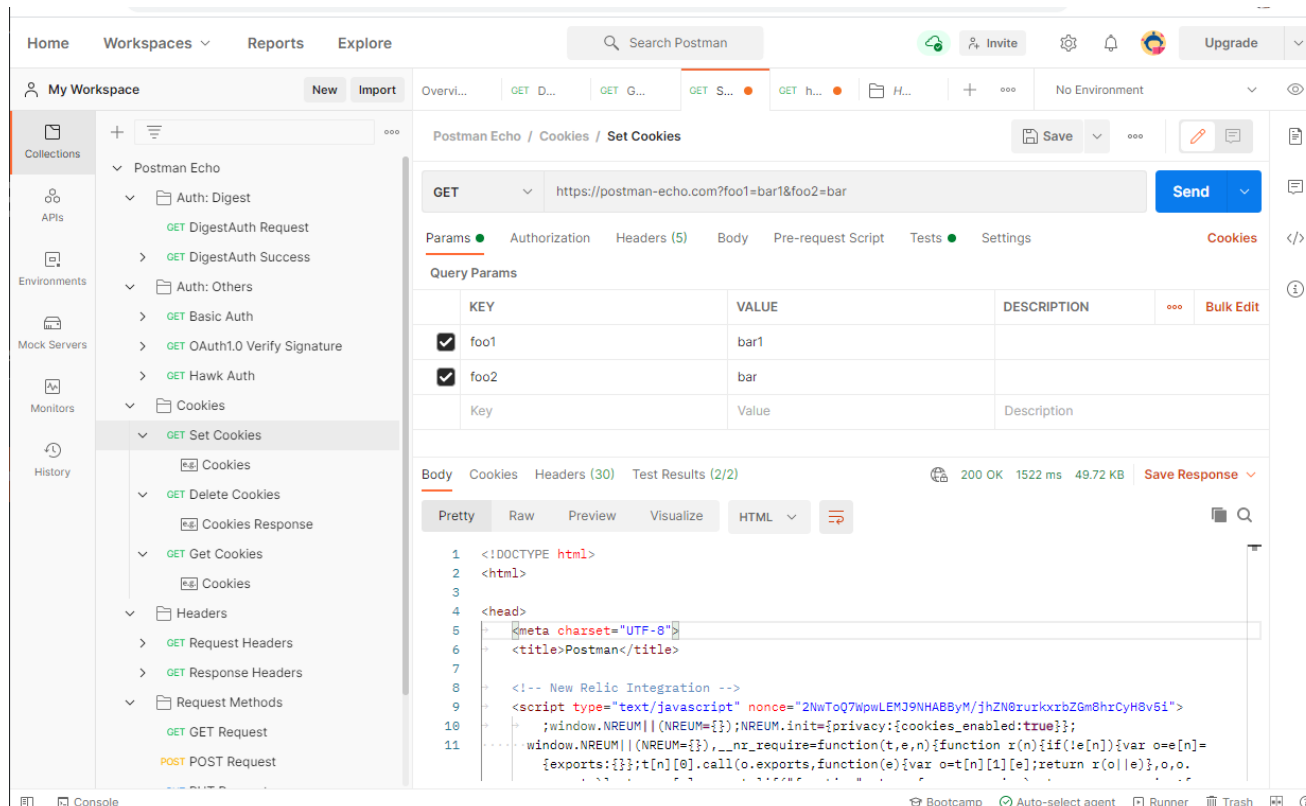


Docker + VM ware

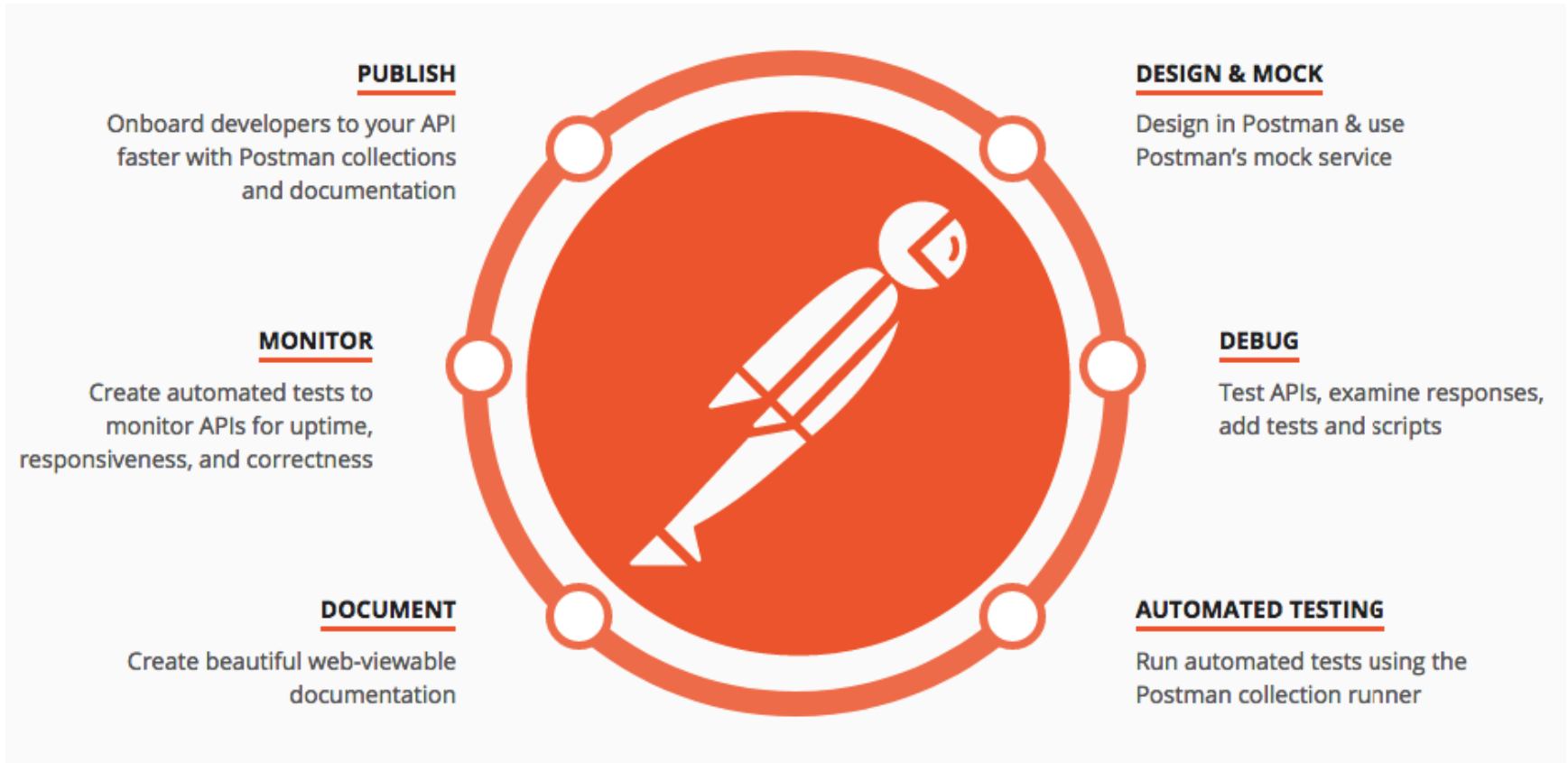


Postman

- Postman is a popular API client that makes it easy for developers to create, share, test and document APIs
- This is done by allowing users to create and save simple and complex HTTP/s requests, as well as read their responses



Postman



FULL STACK DEVELOPER

Full Stack Developer

- A person who can develop both front-end and back-end software.
- Can apply to a web stack, mobile stack, or a native application stack
- Must know how to:
 - Program a front-end software (client side) – GUI, browser
 - Program a back-end software (server side) – server, databases

Font-end Developer

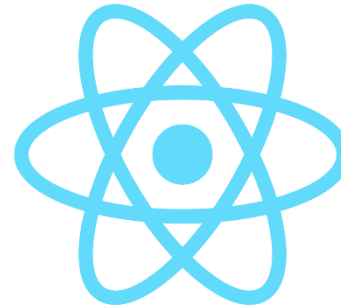
- Builds the visible parts of websites that users see and interact within their web browsers
- The font-end of a website (web or mobile application) is the part a user sees and directly interacts with.
- Basically, the front end is built with languages like:
 - HTML
 - CSS
 - JavaScript

Back-end Developer

- Builds the “under the hood” parts of websites that users don’t interact with directly
- The back end is much more behind-the-scenes and can have some advantages over front end technologies for specific projects
- Back end programming languages include:
 - PHP
 - Java
 - Python

Top Font-End Technologies

- HTML
- CSS
- JavaScript
- Bootstrap
- React
- React Native
- Angular



Top Back-End Technologies

- PHP
- ASP
- Java
- Python
- Node.js
- REST
- C#, C++
- SQL (database)
- MongoDB (database)

